

ACM TECH WEEK

DATA ANALYSIS

ENGLISH PREMIER LEAGUE

Joshika Sri N R
Thiagarajar College of Engineering,
Madurai



DATA SET ANALYSIS

- Dataset contains Premier League match results from 1993 to 2004
- Total features: Season, DateTime, Teams, Scores, Referee, Stats (shots, fouls, cards, etc.)
- Task: Clean the dataset for trend analysis and model training

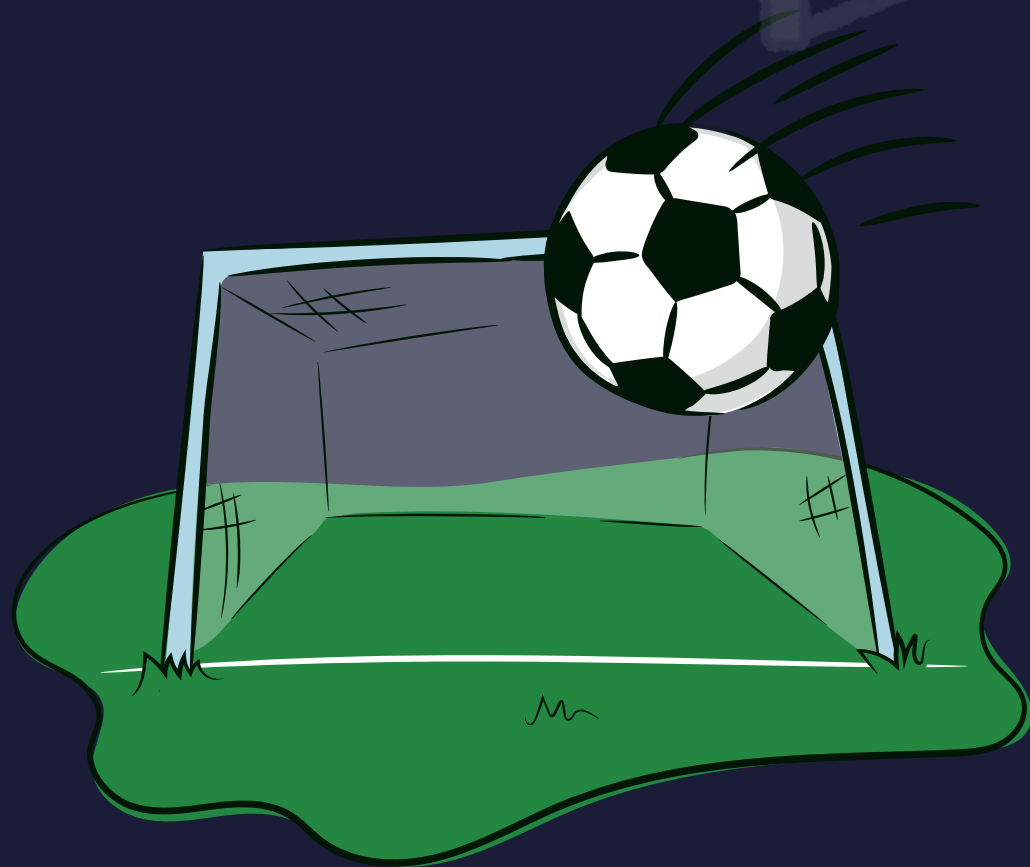
	Season	DateTime	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG
0	1993-94	1993-08-14T00:00:00Z	Arsenal	Coventry	0	3	A	NaN	NaN
1	1993-94	1993-08-14T00:00:00Z	Aston Villa	QPR	4	1	H	NaN	NaN
2	1993-94	1993-08-14T00:00:00Z	Chelsea	Blackburn	1	2	A	NaN	NaN
3	1993-94	1993-08-14T00:00:00Z	Liverpool	Sheffield Weds	2	0	H	NaN	NaN
4	1993-94	1993-08-14T00:00:00Z	Man City	Leeds	1	1	D	NaN	NaN

HTR	...	HST	AST	HC	AC	HF	AF	HY	AY	HR	AR
NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

SAMPLE DATA SET (First Five Rows)



STEP 2



DATA CLEANING - PROCESSES

- Replaced "NA" and empty strings with NaN
- Converted DateTime column to proper datetime format
- Converted numerical columns to correct datatypes
- Filled missing numerical values with median
- Filled missing categorical values with mode (e.g., Referee)
- Dropped rows with missing critical fields (e.g., Team names, scores)
- Removed duplicate rows

Season	DateTime	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG
1993-94	1993-08-14 00:00:00+00:00	Arsenal	Coventry	0	3	A	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Aston Villa	QPR	4	1	H	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Chelsea	Blackburn	1	2	A	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Liverpool	Sheffield Weds	2	0	H	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Man City	Leeds	1	1	D	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Newcastle	Tottenham	0	1	A	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Oldham	Ipswich	0	3	A	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Sheffield United	Swindon	3	1	H	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	Southampton	Everton	0	2	A	NaN	NaN
1993-94	1993-08-14 00:00:00+00:00	West Ham	Wimbledon	0	2	A	NaN	NaN

HTR	Referee	HS	AS	HST	AST	HC	AC	HF	AF	HY	AY	HR	AR
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0
NaN	M Dean	13.0	10.0	6.0	4.0	6.0	4.0	11.0	12.0	1.0	2.0	0.0	0.0

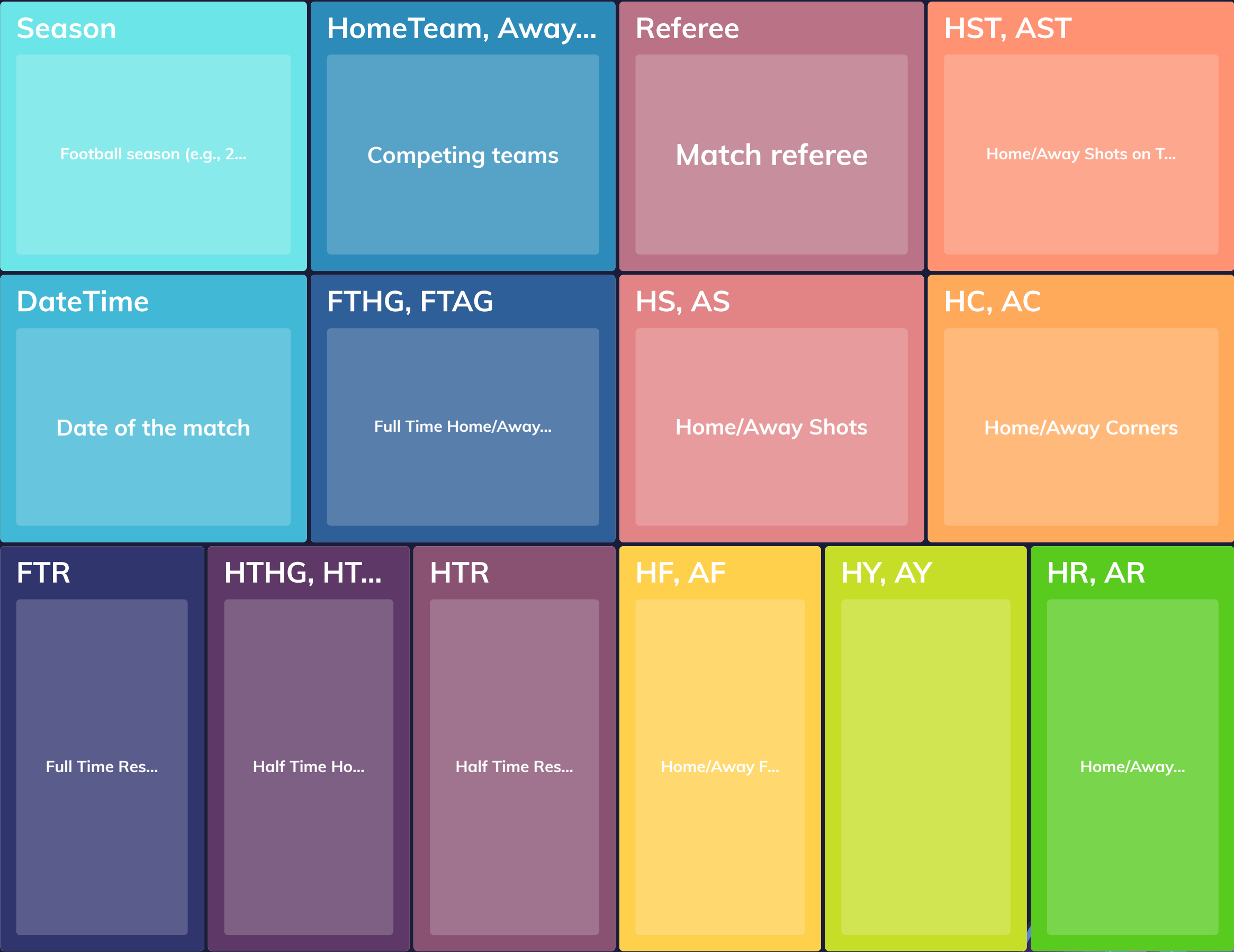
STEP 3

CLEANED DATA PREVIEW & ENCODING

- Final dataset has no major missing or malformed values
- One-hot encoded teams and results for ML-readiness
- Ready for EDA, model training, and trend visualization

Explanation of the Terms involved in the Dataset

Column	Explanation
Season	Football season (e.g., 2003–04)
DateTime	Date of the match
HomeTeam, AwayTeam	Competing teams
FTHG, FTAG	Full Time Home/Away Goals
FTR	Full Time Result (H, D, A)
HTHG, HTAG	Half Time Home/Away Goals
HTR	Half Time Result
Referee	Match referee
HS, AS	Home/Away Shots
HST, AST	Home/Away Shots on Target
HC, AC	Home/Away Corners
HF, AF	Home/Away Fouls
HY, AY	
HR, AR	Home/Away Red Cards

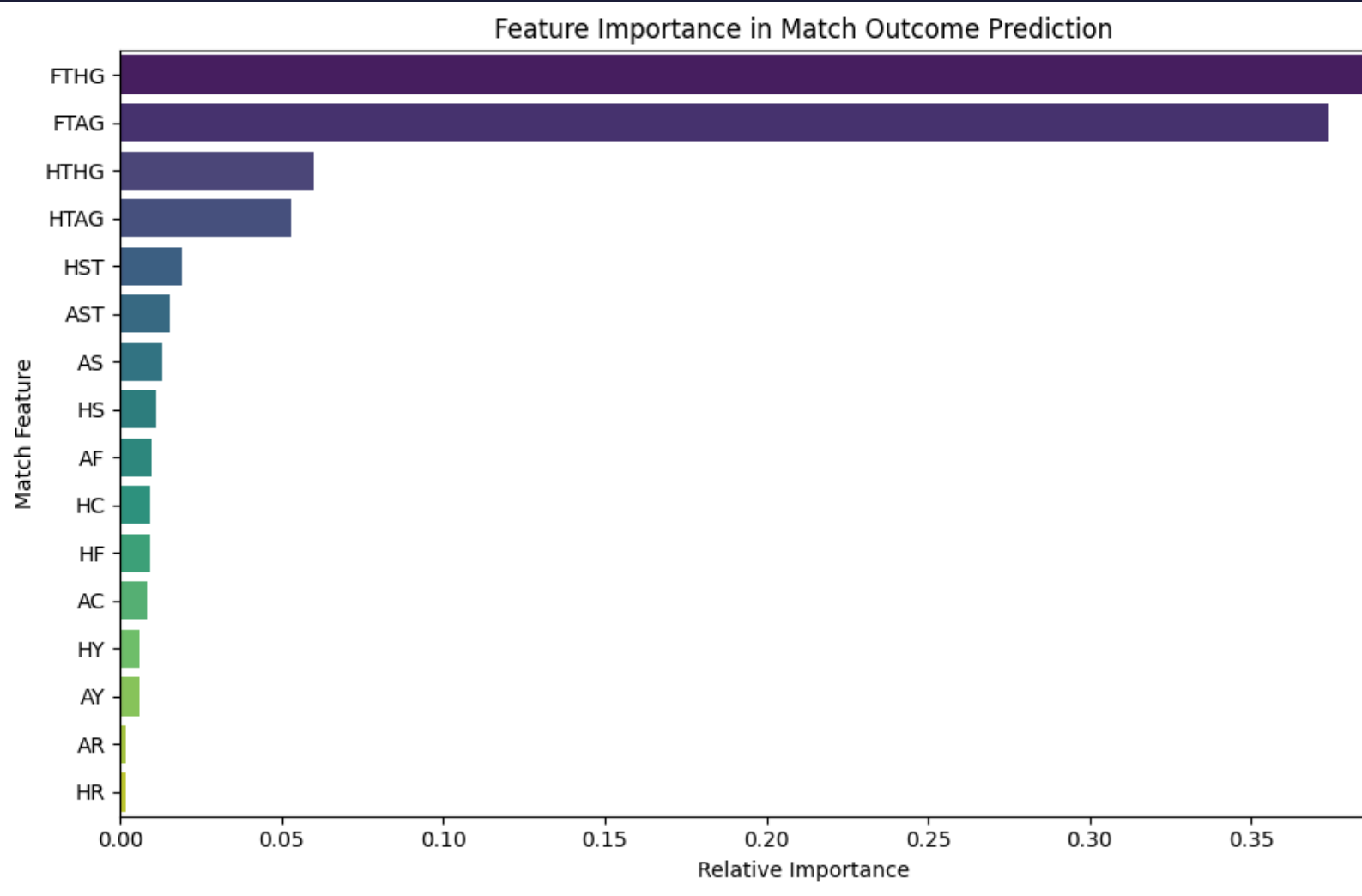




ML MODEL TRAINING - MATCH OUTCOME PREDICTION

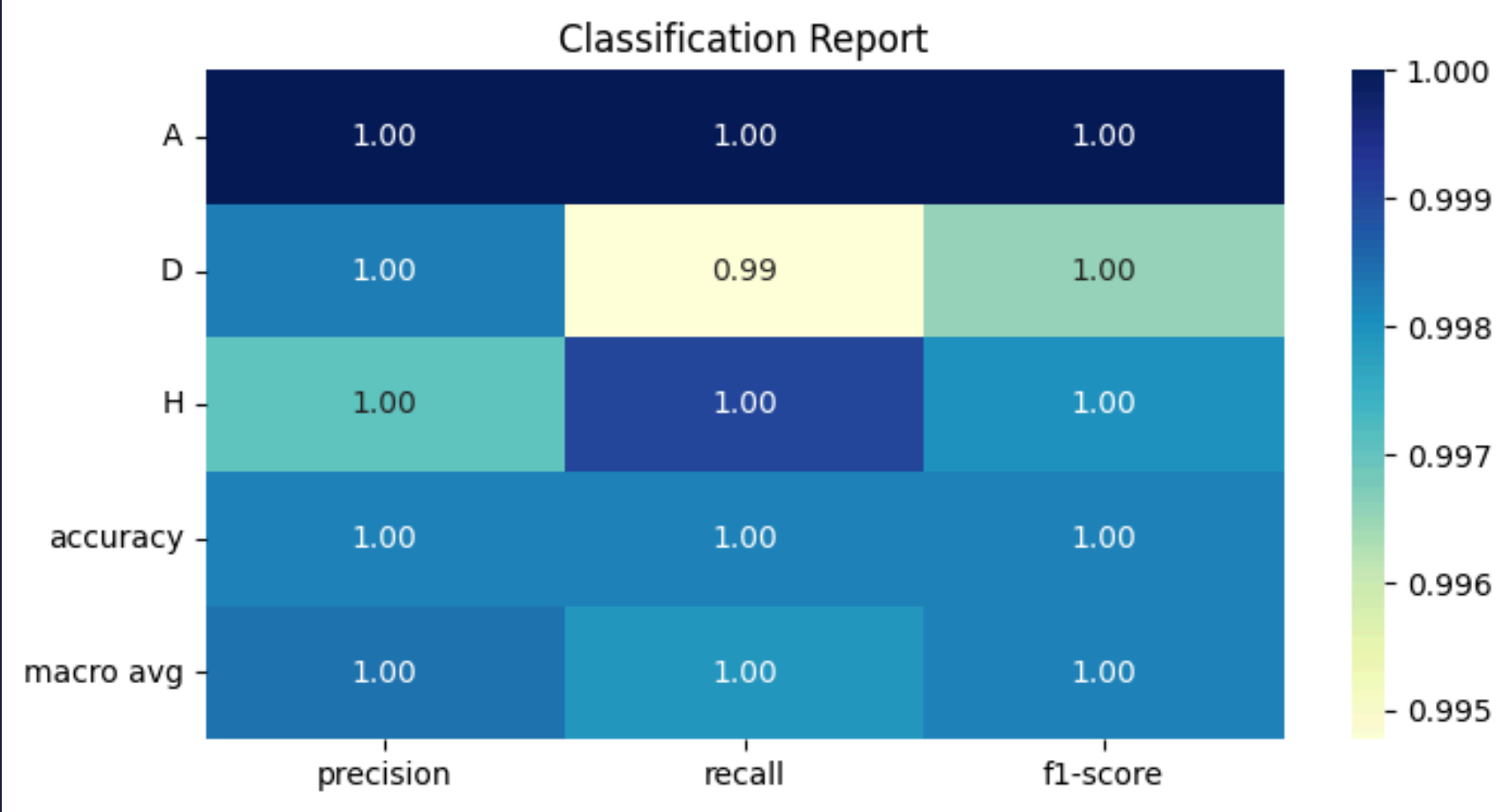
- Trained a Random Forest Classifier to predict match results (Home Win, Draw, Away Win) using match statistics like shots, fouls, and corners.
 - Preprocessed data by removing non-numeric columns and encoding outcomes ($H \rightarrow 0$, $D \rightarrow 1$, $A \rightarrow 2$); used an 80-20 train-test split.
 - Evaluation showed good accuracy for Home Wins, with insights from classification reports and confusion matrix to understand prediction performance.
- 

ML MODEL TRAINING - INSIGHTS



Full-time goals (FTHG, FTAG) are the most influential features in predicting match outcomes, significantly outweighing all other match statistics.

The Random Forest model achieved near-perfect performance with precision, recall, and F1-scores all at or near 1.00 across match outcomes (Home Win, Draw, Away Win), indicating excellent predictive accuracy.



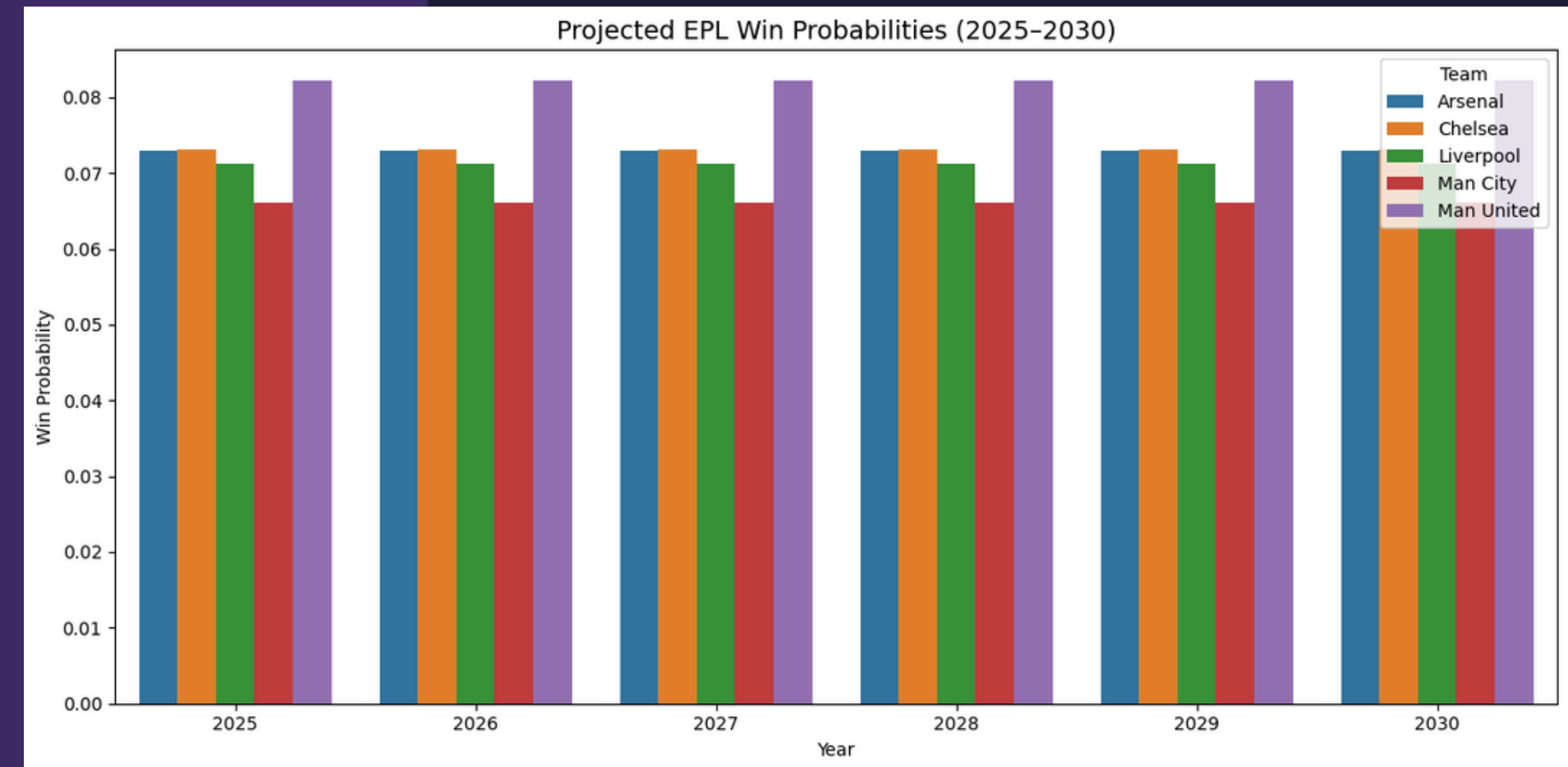
FORECASTING TEAM WINS ACROSS FUTURE SEASONS (2025–2030)

- A heatmap was created to visualize the predicted number of title wins by each top EPL team from 2025 to 2030.
- The X-axis represents the future seasons (2025–2030), and the Y-axis shows the top EPL teams.

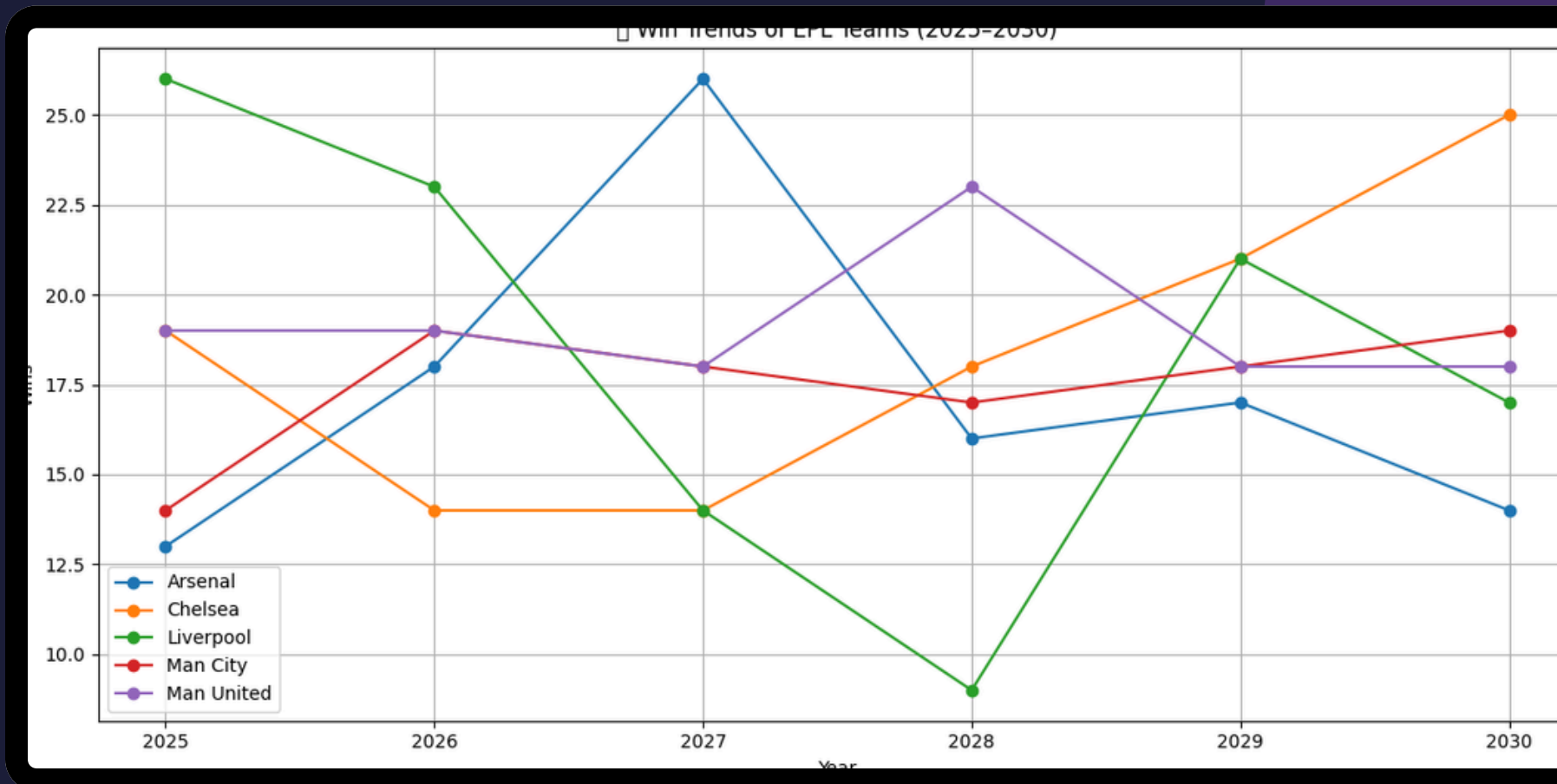


- Heatmap shows predicted title wins (2025–2030) for top EPL teams.
- Darker color = more predicted wins.
- Man United is the strongest title contender across all seasons.
- Arsenal, Chelsea, Liverpool show steady, moderate chances.
- Man City has the lowest win probability among the five

ADVANCED DATA ANALYSIS



WIN TRENDS OF TOP EPL TEAMS (2025–2030)

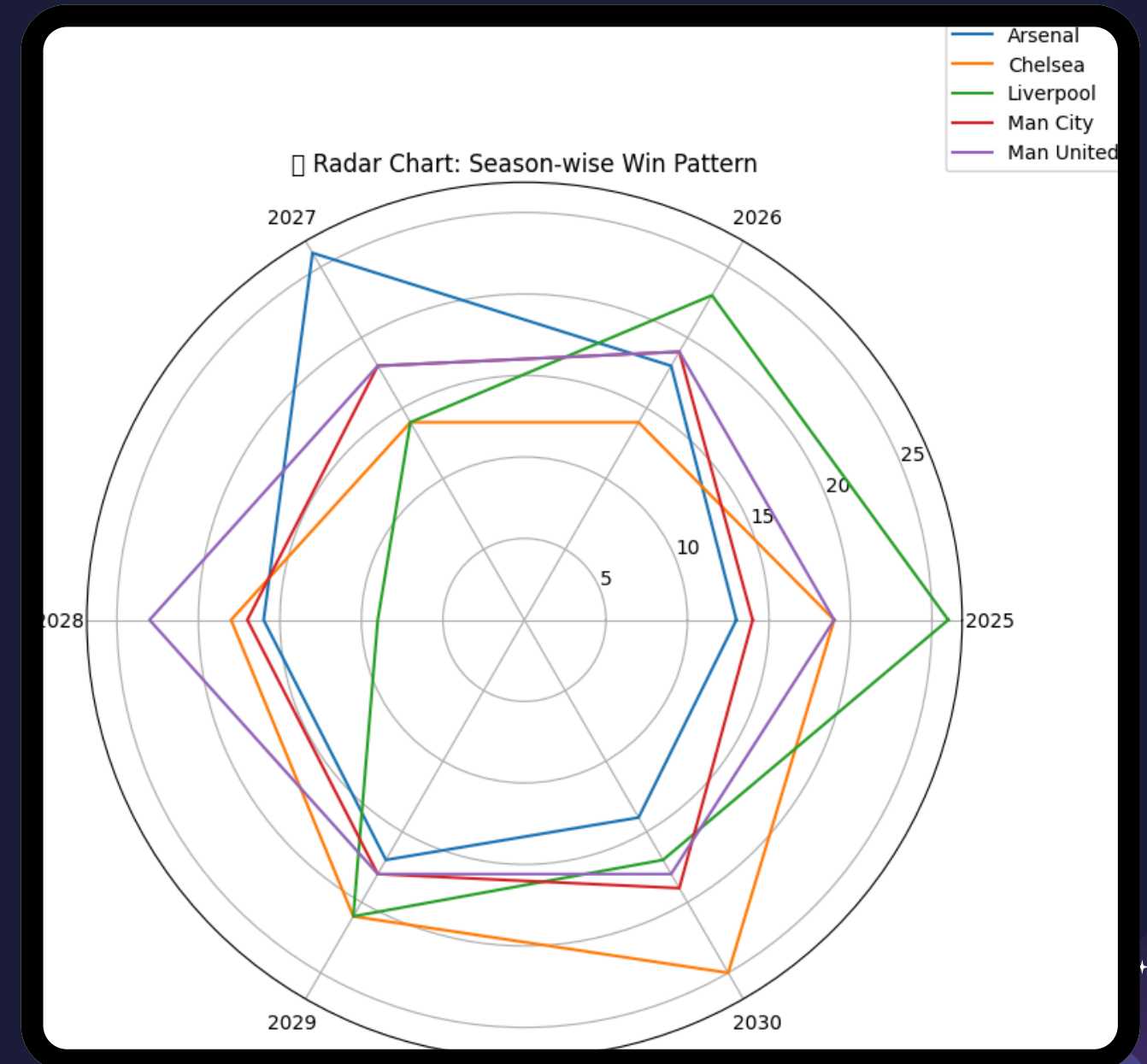


- A line graph was plotted to show predicted number of wins per season for top EPL teams from 2025 to 2030.
- Used trained ML model to estimate future performance trends based on historical data.

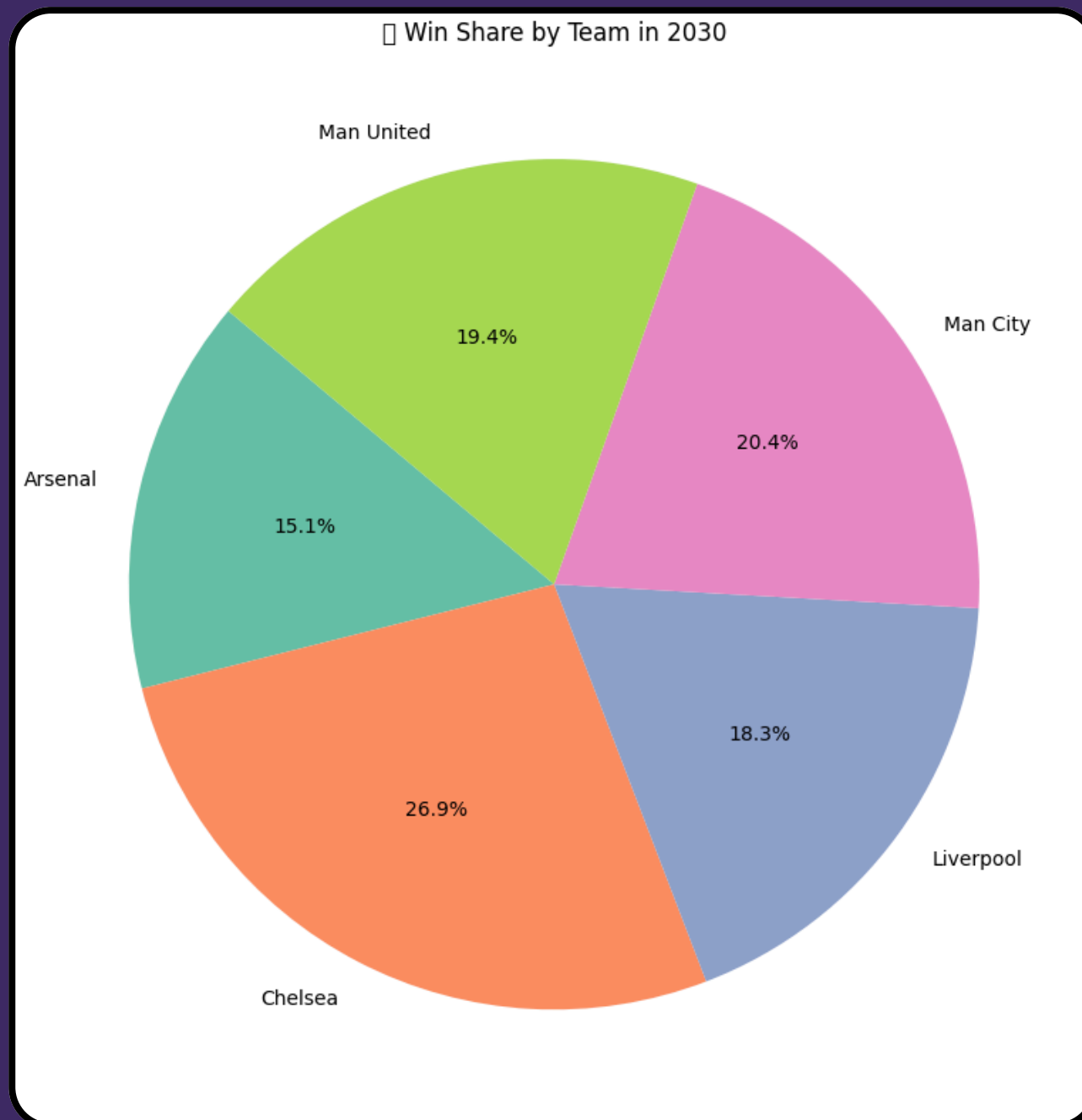
- Chelsea shows the most significant upward trend, potentially becoming dominant by 2030.
- Liverpool starts strong in 2025 but dips mid-period, recovering in 2029.
- Arsenal peaks in 2027 but declines after.
- Man United maintains stable performance, with a peak in 2028.
- Man City remains consistent but with modest win counts compared to others.

SEASON-WISE WIN PATTERN (2025–2030):

- Arsenal peaks in 2027 with highest wins.
- Chelsea shows steady growth, peaks in 2030.
- Liverpool is volatile—strong early, dips in 2028.
- Man City remains consistent across seasons.
- Man United shows stable performance, peaking in 2028.
- Let me know if you'd like it stylized or exported into slide text format.



WIN SHARE BY TEAM IN 2030



- Chelsea leads with 26.9% win share, showing top dominance in 2030.
- Man City (20.4%) and Man United (19.4%) closely follow, indicating strong title contention.
- Liverpool (18.3%) had a moderate season, possibly facing performance dips.
- Arsenal trails with 15.1%, suggesting a possible rebuild or inconsistent form.
- The top 3 teams together hold over 66% of total wins, showing a competitive edge over others.



THANKYOU